Log/Tropical Geometry and the Quotient-Tower Picture

Research-design note for `state_collapser`

May 2026

Status

This is a research-design note, not a settled theorem.

It records a mathematical discussion about whether the `logHRL` quotient-tower construction should be understood not merely as being analogous to tropical geometry, but as a finite, algorithmic reconfiguration of the same logarithmic operation that appears in first-wave tropical geometry.

Attribution

The central conceptual claim in this note is due to the Project Owner.

The Project Owner introduced the stronger thesis that the TeX research document is not merely borrowing tropical language, but may be reconfiguring the entire log/tropical geometry picture in the setting of reinforcement-learning state/action path geometry. The Project Owner also directed the discussion away from later adic/valuation analytic geometry and toward the first-wave coordinatewise logarithmic construction of amoebas and tropical limits.

The core Project Owner question was:

> *Can the logarithm that appears in the path-volume/log-speed-up theorem be the same logarithm that appears in tropical geometry, with the logarithm observed in a different part of the construction?*

The assistant contribution here is organizational: making the dictionary explicit, identifying semiring dequantization as the cleanest bridge, separating strong points from weak points, and writing down theorem-shaped formulations that could guide future mathematical work.

1 Starting Claim

The strong claim under discussion is:

The quotient-tower construction in `logHRL` is not just loosely reminiscent of tropical geometry. It may be a finite, algorithmic version of the same logarithmic simplification that produces tropical geometry.

The guiding phrase that emerged in the discussion is:

loss of fine coordinates while preserving asymptotic decision geometry.

This is the point of contact. Tropical geometry discards fine coordinate data while preserving dominance, order-of-growth, and combinatorial incidence data relevant to the computation. The quotient tower discards fine state/action identity while preserving the outgoing decision geometry relevant to hierarchical control.

2 First-Wave Tropicalization

The relevant tropical picture is the early coordinatewise logarithmic one. One starts with a point in a positive or complex torus and applies a logarithmic map:

$$(z_1, \ldots, z_n) \longmapsto (\log_b |z_1|, \ldots, \log_b |z_n|).$$

For a fixed coordinate value, changing the base alone is only a rescaling:

$$\log_b x = \frac{\log x}{\log b}.$$

So the base change is not interesting by itself. It becomes geometrically meaningful when applied to a family whose coordinates scale with the base:

$$x_b \sim cb^{\alpha}.$$

Then

$$\log_b x_b \to \alpha.$$

The logarithm has discarded the coefficient c and retained the exponent, order-of-growth, or dominance datum α.

This is the relevant mathematical action:

exact coordinate values are forgotten; scale/dominance data are retained.

3 The Amoeba as the Tier-Zero Analogue

One way to place the quotient tower next to this picture is:

> *apply one logarithmic map and obtain an amoeba; pretend this amoeba is the analogue of G_t^0.*

This is not because the amoeba is literally a graph. The point is that the amoeba is still close to the original variety: it is a transformed image that has not yet collapsed all the way to a tropical skeleton.

In the state/action setting, G_t^0 plays a similar role. It is already a discovered, combinatorial representation of the environment, but it is still the finest available tier. It has not yet been coarsened into the quotient tower.

The possible analogy is:

amoeba after one log map	$\longleftrightarrow$	discovered base graph G_t^0
tropical limiting skeleton	$\longleftrightarrow$	coarser tiers G_t^i
full scale-dependent contraction process	$\longleftrightarrow$	quotient tower $G_t^0 \to G_t^1 \to \cdots \to G_t^N$

4 How the Tier Index i Is Like the Logarithmic Base b

The tier index i is not literally the base b.

The better formulation is:

> *i is a discrete event index along a logarithmic visibility flow.*

If one changes b in a raw logarithmic map, then for two fixed positive numbers x, y,

$$\log_b x - \log_b y = \frac{\log x - \log y}{\log b}.$$

As b increases, their distance in log-coordinates shrinks. With infinite precision this is only rescaling. But with a fixed observational threshold, decision resolution, or dominance criterion, more distinctions become invisible as b varies.

Thus, a meaningful dictionary is:

base b	$\longrightarrow$	continuous logarithmic resolution parameter
tier i	$\longrightarrow$	discrete quotient event along that resolution parameter
G_t^i	$\longrightarrow$	graph/decision geometry after distinctions invisible at scale b_i have been contracted
Σ_0^i	$\longrightarrow$	edge list whose distinctions die between scale b_{i-1} and b_i

Equivalently, set

$$\varepsilon = \frac{1}{\log b}.$$

Then increasing b decreases ε, so increasingly fine subscale distinctions disappear. The quotient tower records the finitely many critical moments at which this loss of visibility changes the combinatorial decision structure.

This is the honest version:

> *The tower is not the raw family of maps* $\log_b$*; it is the event-quotient of that family after imposing a finite decision-resolution relation.*

5 The Same Logarithm: Semiring Dequantization

The cleanest way to make the phrase "the same logarithm" precise is semiring dequantization.

Consider

$$\mathrm{Log}_b(x) := \log_b x.$$

This sends multiplication to addition exactly:

$$\log_b(xy) = \log_b x + \log_b y.$$

It sends addition to log-sum-exp:

$$\log_b(x+y) = \log_b\left(b^{\log_b x} + b^{\log_b y}\right).$$

In the limit,

$$\log_b\left(b^{a_1} + \cdots + b^{a_n}\right) \longrightarrow \max_i a_i \qquad (b \to \infty).$$

Thus the logarithm deforms the positive semiring into the max-plus semiring:

ordinary addition	$\longrightarrow$	log-sum-exp $\longrightarrow$ max
ordinary multiplication	$\longrightarrow$	addition

This is the same algebraic move in tropical geometry and in Bellman-style optimal control.

6 Tropical Geometry Side

For a polynomial

$$f(z) = \sum_m c_m z^m,$$

set

$$z_j = b^{x_j}.$$

Then the monomial $c_m z^m$ becomes

$$c_m z^m = c_m b^{\langle m,x \rangle}.$$

After taking logarithms:

$$\log_b |c_m z^m| = \log_b |c_m| + \langle m, x \rangle.$$

The full sum is governed, in the tropical limit, by the dominant term:

$$\log_b \left| \sum_m c_m b^{\langle m,x \rangle} \right| \rightsquigarrow \max_m \left\{ \log_b |c_m| + \langle m, x \rangle \right\}.$$

So algebraic addition of monomials becomes tropical maximum over dominant monomials. The hard algebro-geometric object is replaced by a polyhedral dominance object.

This is why tropical geometry can accelerate or simplify computations such as intersection-theoretic computations: the logarithm moves the problem from fine coordinate geometry to combinatorial dominance geometry.

7 RL / Path Geometry Side

In reinforcement learning or path geometry, the analogous "terms" are not monomials but paths.

Let a trajectory γ have reward, negative cost, or energy $R(\gamma)$, and define a positive path weight

$$W_b(\gamma) = b^{R(\gamma)}.$$

The path partition function from a state s is

$$Z_b(s) = \sum_{\gamma : s \rightsquigarrow *} b^{R(\gamma)}.$$

Applying the same logarithm gives

$$\log_b Z_b(s) = \log_b \left(\sum_{\gamma : s \rightsquigarrow *} b^{R(\gamma)} \right) \longrightarrow \max_{\gamma : s \rightsquigarrow *} R(\gamma).$$

This is the Bellman/tropical operation. The ordinary sum over possible paths degenerates to the optimal path. The same logarithm that tropicalizes sums of monomials also tropicalizes sums over trajectories.

The dictionary is:

tropical geometry	$\longrightarrow$	sum over monomials; $\log_b$; dominant monomial survives
RL/path geometry	$\longrightarrow$	sum over paths/actions; $\log_b$; optimal path/action survives

This is a concrete sense in which the logarithm is the same logarithm.

8 Direct Image Along Quotients

The direct-image discussion is especially important because it explains why max aggregation is not merely an engineering choice.

Let

$$q : S \to \bar{S}$$

be a quotient map. An ordinary pushforward over a fiber might be

$$(q_* F)(\bar{s}) = \sum_{s \in q^{-1}(\bar{s})} F(s).$$

If

$$F(s) = b^{V(s)},$$

then

$$\log_b \left(\sum_{s \in q^{-1}(\bar{s})} b^{V(s)} \right) \longrightarrow \max_{s \in q^{-1}(\bar{s})} V(s).$$

So the tropical direct image is

$$(q_*^{\mathrm{trop}} V)(\bar{s}) = \max_{s \in q^{-1}(\bar{s})} V(s).$$

This matches the PO observation that, for many RL problems, a max direct image is more natural than an average direct image because the coarser state should remember the best available downstream value, not the average value of the fiber.

In this language:

sum aggregation	$\longrightarrow$	ordinary direct image
average aggregation	$\longrightarrow$	normalized/probabilistic direct image
max aggregation	$\longrightarrow$	tropical direct image
log-sum-exp / p-norm families	$\longrightarrow$	dequantizing bridges between these regimes

9 Where the Log Is Observed

The logarithm is visible in different places in the two stories.

In tropical geometry:

> log *first, compute afterward.*

The logarithm is an explicit coordinate transformation:

$$(z_1, \ldots, z_n) \mapsto (\log_b |z_1|, \ldots, \log_b |z_n|).$$

In the quotient-tower RL story:

> *construct the tower, then observe logarithmic search/address complexity.*

The logarithm appears in the complexity theorem and in the Bellman/direct-image algebra. The PO's stronger thesis is that this difference in where the logarithm is observed may not be fundamental. The tower construction itself may be the hidden logarithmic coordinate transformation for path-space geometry.

That is:

> *the quotient tower may be the log/tropicalization map for RL path geometry.*

The log-speed-up would then be the computational shadow of having passed from flat path space to a tropical/combinatorial decision skeleton.

10 Strong Formulation

The strongest formulation currently on the table is:

The quotient tower is a finite, algorithmic log-limit of the state/action path geometry.

Equivalently:

Hierarchical quotient construction is tropicalization before passage to a continuum limit.

Or, in theorem-shaped language:

Given a finite state/action graph with a compatible quotient tower and semiring-valued reward or value data, the dequantized Bellman/direct-image operators along the tower are max-plus pushforwards on the quotient skeletons. If the ordered edge partition is induced by a logarithmic scale or dominance filtration, then the quotient tower is the finite combinatorial skeleton of the associated path-space tropicalization.

This is not yet a theorem. It indicates the structure one would need to define in order to make the statement theorem-grade.

11 What Must Be Added To Make This More Than Analogy

The weak point is that the current quotient tower can be built from an arbitrary ordered edge partition:

$$\mathcal{A}_t^0 = \Sigma_0^1 \sqcup \cdots \sqcup \Sigma_0^d.$$

An arbitrary ordered partition is not automatically a logarithmic scale filtration.

To make the tropical/log identification honest, one needs extra structure, such as:

- an edge/action distinguishability scale;
- a reward/value dominance scale;
- a path-volume contribution scale;
- a decision-resolution threshold;
- a valuation-like filtration on actions or path families;
- a semiring-valued path partition function whose dequantization induces the partition.

For example, one could seek a scale function

$$\lambda : \mathcal{A}_t^0 \to \mathbb{R}$$

such that

$$e \in \Sigma_0^i \quad \Longleftrightarrow \quad \lambda(e) \in [\lambda_i, \lambda_{i+1}).$$

Then i becomes a discrete critical value of the logarithmic filtration, and the tier construction is much closer to a literal tropicalization process.

Another possible route is to define an indistinguishability relation depending on a logarithmic resolution parameter:

$$s \sim_\varepsilon s' \quad \Longleftrightarrow \quad \text{the outgoing decision data of } s, s' \text{ differ only below resolution } \varepsilon.$$

Then the tower could be recovered at critical values

$$\varepsilon_i = \frac{1}{\log b_i}.$$

This would make precise the slogan:

> *i is a discrete scale index for the log-base flow.*

12 Possible Commutative Diagram

A future theorem might be organized around a diagram of this form:

$$\begin{array}{ccc} \text{positive path weights on } G_t^0 & \xrightarrow{q_*} & \text{positive fiber/path aggregates on } G_t^i \\ \downarrow \log_b & & \downarrow \log_b \\ \text{dequantized values on } G_t^0 & \xrightarrow{q_*^{\text{trop}}} & \text{max-plus fiber/path values on } G_t^i \end{array}$$

In the limit $b \to \infty$, the bottom horizontal map should be the tropical direct image:

$$(q_*^{\text{trop}} V)(\bar{s}) = \max_{s \in q^{-1}(\bar{s})} V(s).$$

The question is whether the quotient tower itself can also be recovered from the same dequantizing scale data, rather than merely being a useful structure on which the dequantized operators act.

13 Strong Points Of The Analogy

The touch points that currently look mathematically serious are:

- Both stories replace fine coordinate geometry by scale/dominance geometry.

- Both stories turn sums into maxima through the same logarithmic limiting operation.
- Both stories produce combinatorial skeletons that preserve the data relevant to the computation.
- Both stories simplify hard computations by moving from flat/fine geometry to structured quotient or polyhedral geometry.
- The RL Bellman operator is naturally max-plus, which is precisely the tropical semiring.
- Max direct image along quotient fibers is exactly the tropicalization of sum direct image.
- The tier index i can plausibly be interpreted as a discrete critical index along a logarithmic scale filtration.

14 Weak Points Of The Analogy

The main weak points are:

- Changing b alone only rescales a fixed logarithmic image.
- A quotient tower built from an arbitrary edge partition is not automatically a tropical object.
- Tropical geometry usually begins with algebraic equations, monomials, and coordinates; the RL tower begins with a discovered transition graph and contraction policy.
- The current package architecture does not yet define the scale/valuation structure that would force the ordered edge partition to be logarithmic.
- The tower currently encodes decision equivalence, not necessarily order-of-magnitude equivalence, unless those are identified by an additional theorem or construction.

These weak points do not defeat the claim. They specify what has to be built or proved.

15 Research Tasks Suggested By This Discussion

The immediate mathematical tasks are:

1. Define a scale or dominance filtration on state/action graphs that can generate the ordered edge partition used by the tower.
2. Define path partition functions $Z_b(s)$ and quotient fiber partition functions compatible with tower morphisms.
3. Prove that $\log_b$ dequantization turns ordinary path/fiber aggregation into max-plus Bellman and direct-image aggregation.
4. Determine whether the tower partitions can be recovered as critical equivalence classes of a logarithmic visibility relation.

5. Clarify when the quotient tower is a tropical skeleton of path geometry rather than merely a useful hierarchical abstraction.

6. Identify the exact hypotheses under which the log-speed-up theorem and tropical dequantization theorem become the same statement viewed from different sides.

16 Provisional Bottom Line

The conservative statement is:

> *The same logarithmic semiring dequantization that underlies first-wave tropical geometry also underlies max-plus Bellman aggregation and max direct images along quotient fibers.*

The stronger Project Owner thesis is:

> *The state/action quotient tower is itself the finite algorithmic analogue of the tropical/logarithmic limiting process, with the tier index recording discrete critical events in a logarithmic visibility flow.*

That stronger thesis is not established here, but it is now framed in a way that looks mathematically attackable rather than merely metaphorical.

17 Relation To Semiring-Valued Polynomial-Span Dataflow

The Project Owner's reading is that the relevant Dudzik–Velickovic line of work is structurally tropical, even when it is not advertised in those words. That is: Dudzik is, in this sense, secretly operating as a tropical geometer.

This should not be stated as a claim about the authors' intent. The point is structural. Dudzik–Velickovic foreground graph neural networks, dynamic programming, polynomial spans, pullback, pushforward, bag/list monads, semirings, and algorithmic alignment. But the tropical geometry is already sitting inside the semiring choices.

The same point applies to the Gavranovic–Lessard–Dudzik–von Glehn–Araujo– Velickovic categorical deep learning paper: the paper presents an algebraic architecture story, but the moment the architecture is evaluated over tropical or idempotent semirings, it is also a tropical-geometric story in disguise.

Their basic graph-dataflow template has the form

$$[W, R] \xrightarrow{i^*} [X, R] \xrightarrow{p_\otimes} [Y, R] \xrightarrow{o_\oplus} [Z, R].$$

Here R is the value or latent space, and the two aggregation operations $\otimes$ and $\oplus$ supply the algebra needed to combine message arguments and aggregate incoming messages. Their key observation is that the same polynomial-span/integral-transform skeleton can describe both ordinary GNN message passing and dynamic programming, once one chooses the appropriate coefficient algebra.

On the dynamic-programming side, one naturally meets tropical semirings such as

$$(\mathbb{N} \cup \{\infty\}, \min, +)$$

or sign-reversed max-plus variants. This is not an incidental coefficient choice. It is the Bellman/tropical algebra of optimal path computation.

The present note adds the following extra interpretation. Dudzik–Velickovic fix the graph/dataflow diagram, vary the semiring or latent value algebra, and observe that GNNs and dynamic programming share the same abstract polynomial-span computation. The present quotient-tower perspective treats the passage between ordinary positive aggregation and tropical Bellman aggregation as a logarithmic degeneration, views max-plus direct image as the tropicalization of sum direct image, and asks whether the quotient tower itself is the finite tropical skeleton of state/action path geometry.

So the connection is not merely that both stories mention semirings. The shared structure is:

$$\text{pullback} \quad + \quad \text{product-like local combination} \quad + \quad \text{sum/max-like pushforward aggregation.}$$

In Dudzik–Velickovic, this appears as polynomial-span dataflow. In the log/tropical quotient-tower picture, the same architecture is read through the dequantizing map

$$\log_b \left(\sum_i b^{a_i} \right) \longrightarrow \max_i a_i.$$

Thus a sum-type pushforward over a fiber,

$$(q_* F_b)(\bar{s}) = \sum_{s \in q^{-1}(\bar{s})} b^{V(s)},$$

dequantizes to the tropical direct image

$$(q_*^{\mathrm{trop}} V)(\bar{s}) = \max_{s \in q^{-1}(\bar{s})} V(s).$$

This is exactly the state/action quotient analogue of the message pushforward $o_\oplus$, with $\oplus$ specialized to a tropical aggregator.

The rhetorical point should be made carefully:

> *Dudzik–Velickovic identify the semiring-polynomial-span form of graph message passing and dynamic programming. The present perspective reads the tropical semiring appearing in their dynamic-programming examples not as an isolated coefficient choice, but as the logarithmic degeneration of ordinary positive aggregation. This moves the semiring parameter itself into the geometry.*

The additional `state_collapser` contribution is the tower:

$$G_t^0 \twoheadrightarrow G_t^1 \twoheadrightarrow \cdots \twoheadrightarrow G_t^N.$$

The question is no longer only how to run semiring-valued dataflow on one graph. The question is how semiring-valued dataflow behaves under quotient maps, direct images, lifts, frozen tiers, and scale-indexed contraction events. This is where the Project Owner's stronger log/tropical thesis genuinely goes beyond the Dudzik–Velickovic setup.

18 Literature Trail For This Connection

The immediate machine-learning/categorical-dataflow literature is:

- Dudzik and Velickovic, *Graph Neural Networks are Dynamic Programmers*. This is the closest source. It identifies a polynomial-span/integral-transform skeleton common to GNN message passing and dynamic programming, with semiring structure on the latent/value space R. The OpenReview page is here.
- Gavranovic, Lessard, Dudzik, von Glehn, Araujo, and Velickovic, *Position: Categorical Deep Learning is an Algebraic Theory of All Architectures*. This places the Dudzik-style algebraic architecture perspective inside a broader categorical deep learning program. The arXiv page is here, and the project page is here.
- Dudzik, *Quantales and Hyperstructures: Monads, Mo' Problems*. This is useful background for why the semiring/quantale/hyperstructure thread in Dudzik's work is not accidental.

The older dynamic-programming and path-problem background is:

- de Moor, *Categories, Relations and Dynamic Programming*. This is one of the categorical dynamic-programming sources explicitly used by Dudzik–Velickovic.
- Carre, *Semirings and path spaces*. This is a classical algebraic path-problem source: graph path problems are treated through semiring-like algebraic structure.
- Baras and Theodorakopoulos, *Path Problems in Networks*. This is a book-length treatment of the algebraic path problem and its applications.
- Master, *The Open Algebraic Path Problem*. This extends algebraic path problems to open networks and emphasizes functoriality.
- GraphBLAS. This is the engineering-standard version of the same semiring insight: graph algorithms can be expressed as linear-algebraic operations over chosen semirings.

The polynomial-span/polynomial-functor background is:

- Gambino and Kock, *Polynomial functors and polynomial monads*. This is a foundational source for polynomial functors and polynomial monads.

- Weber, *Polynomials in categories with pullbacks*. This is useful for the more general categorical setting of polynomial functors beyond locally cartesian closed categories.
- Willerton, *Integral Transforms and the Pull-Push Perspective, I*. This is helpful conceptual background for the pull-push/integral-transform language.
- Tambara, *On multiplicative transfer*. This is relevant because finite polynomial diagrams and multiplicative/additive transfer are part of the deeper Lawvere-theory story around commutative semirings.

The tropical and semiring-scheme background is:

- Maclagan and Sturmfels, *Introduction to Tropical Geometry*. This is the standard entry point for tropical geometry, tropical varieties, and the algebraic/combinatorial role of the tropical semiring.
- Giansiracusa and Giansiracusa, *Equations of tropical varieties*. This is especially relevant because it treats tropical hypersurfaces through semiring schemes over idempotent semirings.
- Maclagan, *Introduction to tropical algebraic geometry*. This is a shorter expository bridge into tropical algebraic geometry.

The key interpretive bridge is:

> *semiring-valued graph dataflow* $\longrightarrow$ *polynomial-span/pull-push architecture* $\longrightarrow$ *dynamic programming over tropical coefficients* $\longrightarrow$ *logarithmic dequantization explains why tropical coefficients are the natural Bellman limit* $\longrightarrow$ *quotient towers ask how this tropicalized dataflow behaves under scale-indexed contraction and lifting.*

19 Scale Filtration Formulation

The preceding analogy should be stated in a way that does not over-identify the constructions. The common object is not "the logarithm" as a single function. The common object is a filtration by scale.

> *Logarithms, valuations, degeneration parameters, tropical limits, special fibers, and quotient towers are different ways of extracting scale filtration data from a complicated object, computing on the resulting reduced, graded, or skeletal object, and then lifting local information back through the fibers.*

In the Archimedean setting, this appears as coordinatewise log maps and amoebas. In tropical geometry, it appears as valuation or leading-order data. In adic and degeneration settings, it appears as a parameter whose powers filter the object and whose special fiber records the reduced scale-visible geometry. In each case, the logarithmic operation is not merely a numerical change of coordinates. It is a mechanism for passing from a flat fine object to a filtered object, and then to a reduced, skeletal, or associated-graded shadow on which computation is cheaper.

The `state_collapser` analogy is therefore not that the state/action graph is literally tropicalized or adified. Rather, the quotient tower is a finite algorithmic analogue of this pattern. The state/action graph is equipped with an ordered contraction structure that behaves like a scale filtration. Passing to higher tiers discards fine distinctions while preserving the decision data visible at that scale. Computation can then be performed on the coarser quotient object, and executable behavior is recovered by lifting through the state/action fibers.

The broad paradigm is: *fine object → scale filtration/contraction → reduced or skeletal computation → local lift back through fibers.*

The logarithm is the signature of this passage from flat fine geometry to scale-organized computation. Sometimes it appears as an actual coordinatewise logarithm. Sometimes it appears as valuation or order-of-magnitude data. Sometimes it appears as a degeneration parameter. Sometimes it appears as the source of max-plus/tropical algebra. In the `logHRL` theorem, it appears as the observed complexity reduction produced by hierarchical address structure.

The hidden mathematical word is therefore *filtration*. The logarithm is how multiplicative or analytic size becomes additive filtration data. The tower is not yet proved to be a tropicalization, but it is plausibly a finite computational filtration whose quotient tiers play the role that associated graded objects, special fibers, skeleta, or tropical shadows play elsewhere.

Tyler: Here's a picture I drew that, to me, shows "this same sort of structure we invented for log speed-up weirdly shows up in *other* logarithmic settings as well, where it serves a very similar purpose":

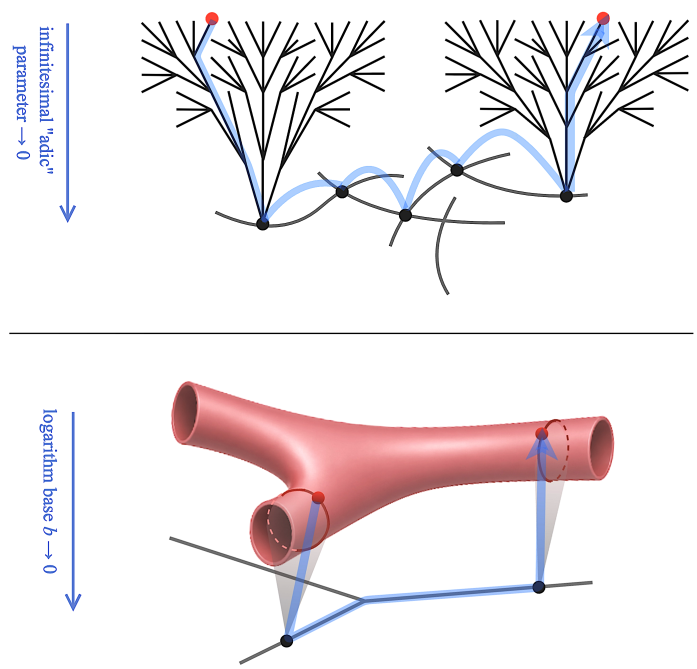

Figure 1: (Top) Non-Archimedean analytic, or "*adic*" geometry in as a curve collapses into a logarithmic pole. (Bottom) The degeneration of an amoeba down to a combinatorial curve as the base of its defining logarithm approaches 0.